Linear Models for Regression and Classification

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Regression

- Regression is a task for modeling the dependence of a response y on a covariate $\mathbf{x}$.
- Regression model can be described by

$$y = f(\mathbf{x}) + \epsilon$$

  where ϵ is an observation noise.
- Depending on $f(\mathbf{x})$,
  - Linear regression: $f(\mathbf{x}) = \sum_{j=1}^{M} w_j \phi_j(\mathbf{x}) + w_0$
  - Kernel regression: $f(\mathbf{x}) = \sum_{i=1}^{N} w_i \phi(\mathbf{x}, \mathbf{x}_i) + w_0$

  where $\phi_j(\cdot)$ is a basis function and $k(\cdot, \cdot)$ is a kernel.

Linear Regression

- Linear regression is a linear model over basis functions $\phi(\mathbf{x})$:

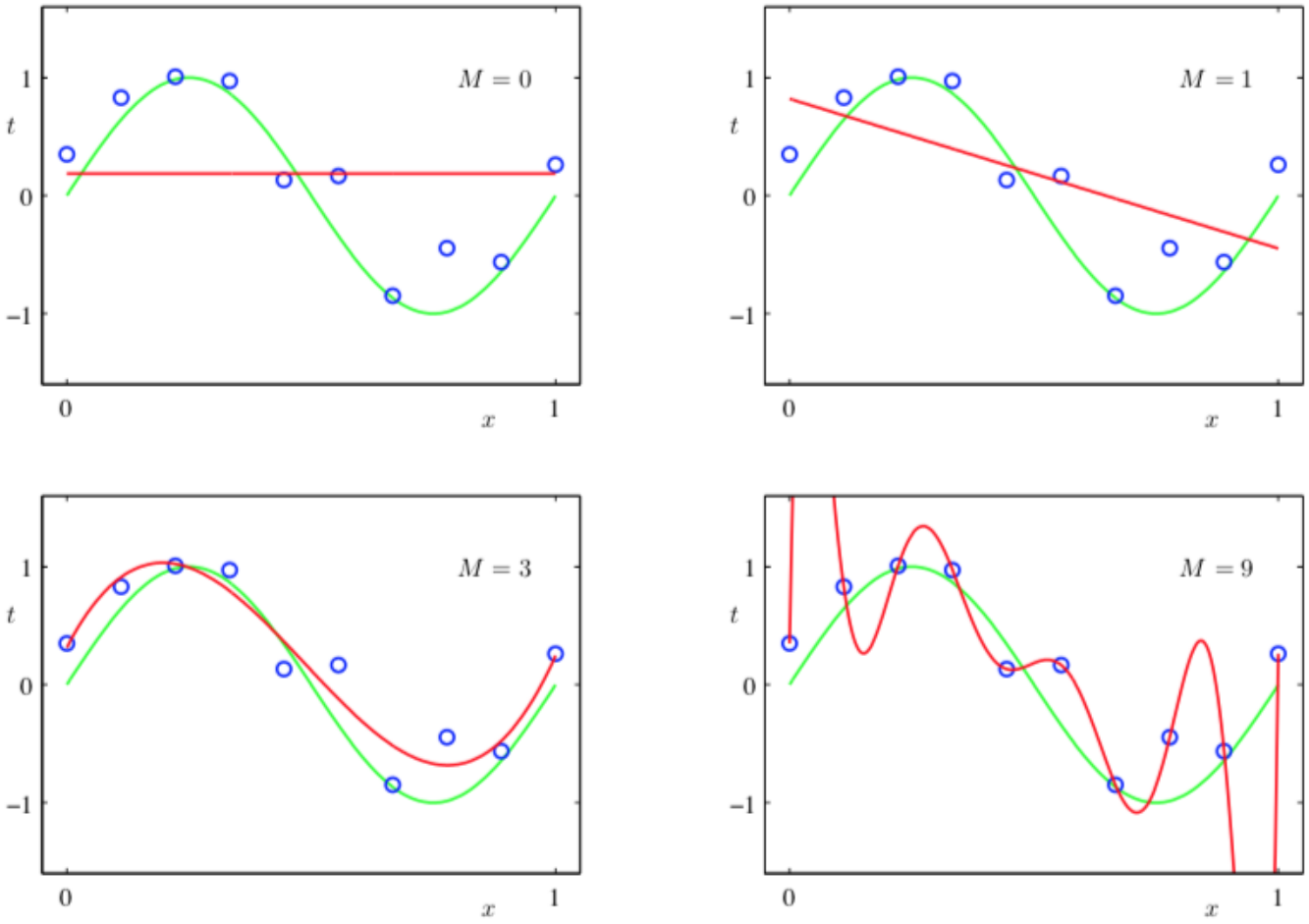
$$f(\mathbf{x}) = \sum_{j=0}^{M} w_j \phi_j(\mathbf{x}) = \mathbf{w}^\top \phi(\mathbf{x})$$

where $\phi_j(\cdot)$ is a basis function and

$$\mathbf{w} = [w_0, w_1, \ldots, w_M]^\top$$
$$\phi(\cdot) = [\phi_0(\cdot), \phi_1(\cdot), \ldots, \phi_M(\cdot)]^\top .$$

Polynomial Regression

- $y_t = \sum_{j=0}^{M} w_j \phi_j(x_t) = \sum_{j=0}^{M} w_j x_t^j.$



Basis Functions

- Polynomial function: $\phi_j(x) = x^j$.
- Gaussian basis function: $\phi_j(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}-\boldsymbol{\mu}_j\|^2}{2\boldsymbol{\sigma}^2}\right)$.
- Spline basis function, sigmoidal basis function, hyperbolic tangent basis function, Fourier basis function, wavelet basis function, and etc.

Least Squares Method

Least Squares Method

- Given a training dataset $\{(\mathbf{x}_t, y_t)\}_{t=1}^{N}$, we can determine the weight vector $\mathbf{w}$, minimizing

$$\mathcal{L}_{\mathrm{LS}} = \frac{1}{2}\sum_{t=1}^{N}\left(y_t - \mathbf{w}^\top \phi(\mathbf{x}_t)\right) = \frac{1}{2}\|\mathbf{y} - \mathbf{\Phi}\mathbf{w}\|^2$$

  where $\mathbf{y} = [y_1, \ldots, y_N]^\top$ and $\mathbf{\Phi} \in \mathbb{R}^{N\times(M+1)}$ is a design matrix with $\Phi_{tj} = \phi_j(\mathbf{x}_t)$.

- The design matrix can be defined as

$$\mathbf{\Phi} = \begin{bmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_M(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_M(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_M(\mathbf{x}_N) \end{bmatrix}.$$

Least Squares Method

- To find a local solution of $\mathcal{L}_{\mathrm{LS}}$, $\frac{\partial \mathcal{L}_{\mathrm{LS}}}{\partial \mathbf{w}} = 0$ should be computed:

$$\frac{\partial \mathcal{L}_{\mathrm{LS}}}{\partial \mathbf{w}} = \mathbf{\Phi}^\top \mathbf{y} - \mathbf{\Phi}^\top \mathbf{\Phi} \mathbf{w} = 0$$
$$\mathbf{\Phi}^\top \mathbf{\Phi} \mathbf{w} = \mathbf{\Phi}^\top \mathbf{y}.$$

- Thus, least squares estimate of $\mathbf{w}$ is given by

$$\mathbf{w}_{\mathrm{LS}} = (\mathbf{\Phi}^\top \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \mathbf{y} = \mathbf{\Phi}^\dagger \mathbf{y}$$

where $\mathbf{\Phi}^\dagger$ is the Moore-Penrose pseudo-inverse and $\mathbf{\Phi}^\top \mathbf{\Phi}$ is known as the Gram matrix.

Ridge Regression

Ridge Regression

- Compared to least squares method, ridge regression additionally considers a Euclidean norm-based regularizer:

$$\mathcal{L}_{\text{ridge}} = \frac{1}{2}\|\mathbf{y} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

  where λ is a hyperparameter for the regularizer.
- Solving $\frac{\partial \mathcal{L}_{\text{ridge}}}{\partial \mathbf{w}} = 0$ leads to

$$\mathbf{w}_{\text{ridge}} = (\lambda \mathbf{I} + \mathbf{\Phi}^\top \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \mathbf{y}.$$

Least Mean Square

Least Mean Square

- Least mean square is a gradient descent method which minimizes the instantaneous error $\mathcal{L}_t$, where

$$\mathcal{L}_{\mathrm{LS}} = \sum_{t=1}^{N} \mathcal{L}_t = \frac{1}{2} \sum_{t=1}^{N} \left(y_t - \mathbf{w}^\top \phi(\mathbf{x}_t) \right)^2 .$$

- The gradient descent method leads to the updating rule for $\mathbf{w}$ that is of the form

$$\begin{aligned} \mathbf{w} &\leftarrow \mathbf{w} - \eta \nabla \mathcal{L}_t \\ &\leftarrow \mathbf{w} + \eta \left(y_t - \mathbf{w}^\top \phi(\mathbf{x}_t) \right) \phi(\mathbf{x}_t) \end{aligned}$$

where $\eta > 0$ is a hyperparameter, referred to as learning rate.

Linear Classification

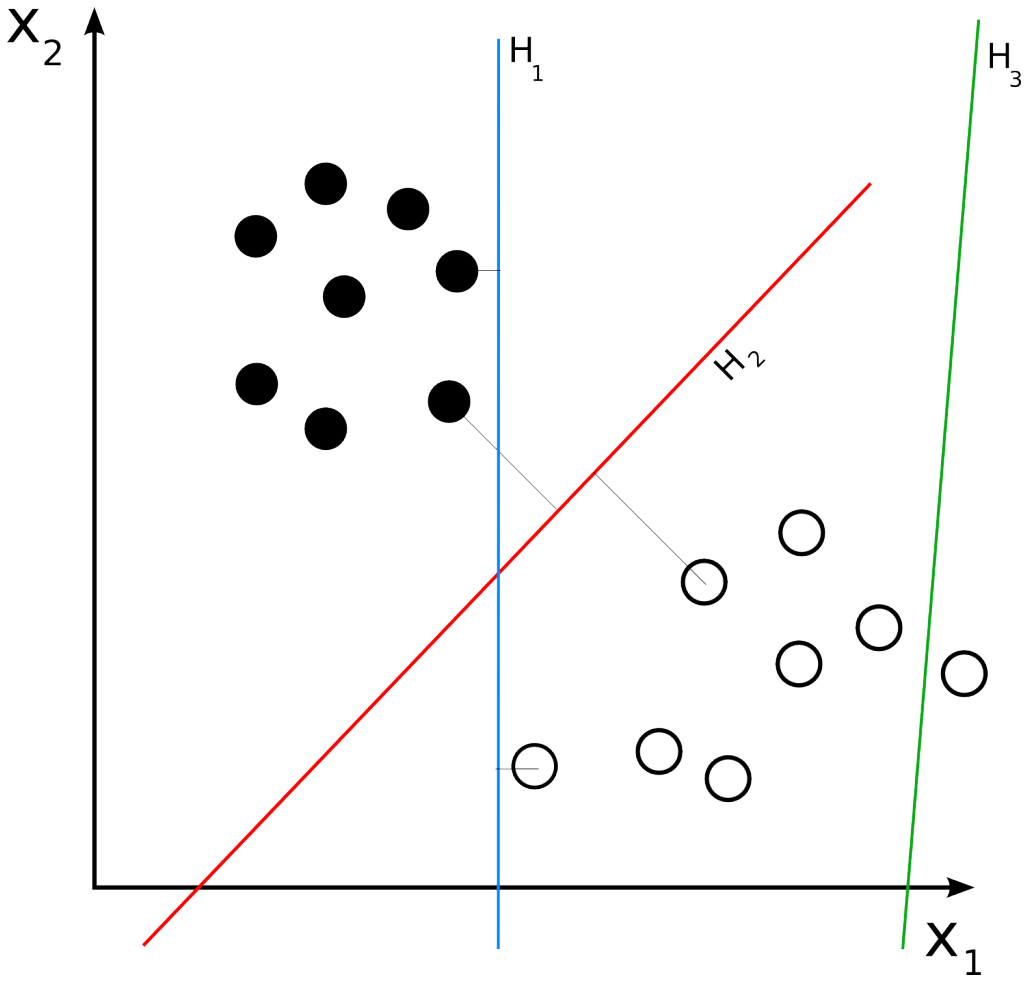
Classification

- Classification is a task for modeling the dependence of a binary-valued (or discrete-valued) y on a covariate $\mathbf{x}$.
- Binary classification model can be written as

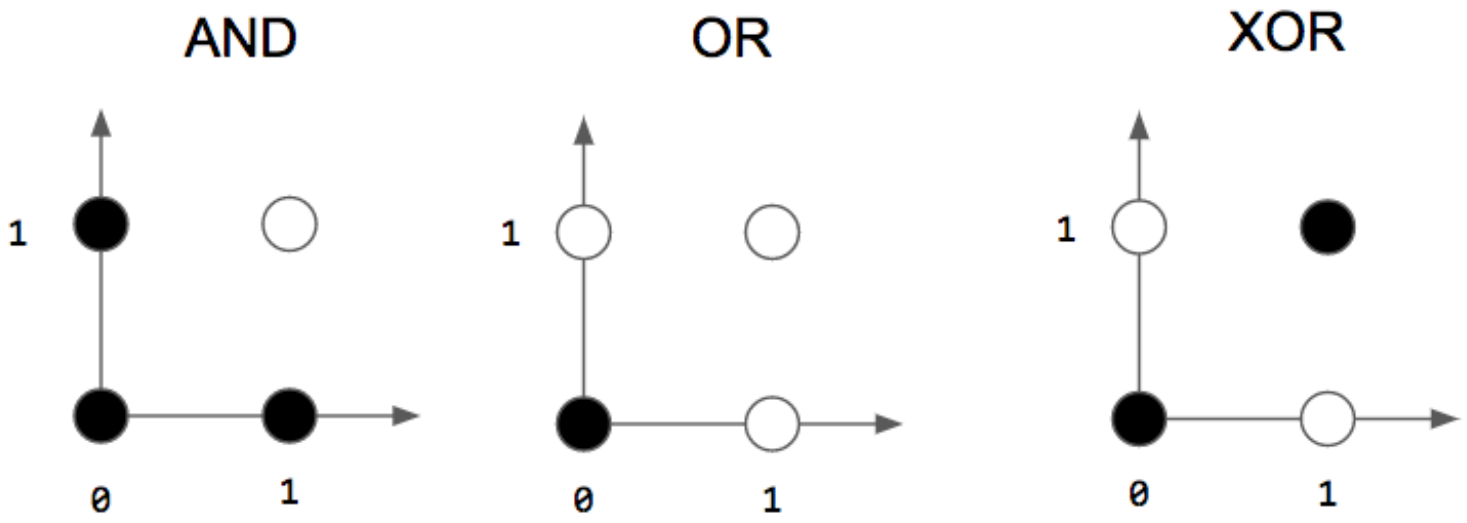
$$f(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{w}^\top \phi(\mathbf{x}) > t \\ 0 & \text{otherwise} \end{cases}$$

where t is a threshold.

Linear Classification



Linearly Separability



Logistic Regression

Logistic Regression

- Predict a binary output $y_n \in \{0, 1\}$ from an input $\mathbf{x}_n$.
- The logistic regression models the input-output by a conditional Bernoulli distribution

$$\mathbb{E}[y_n|\mathbf{x}_n] = \mathbb{P}(y_n = 1|\mathbf{x}_n) = \sigma(\mathbf{w}^\top \mathbf{x}_n),$$

where

$$\sigma(\xi) = \frac{1}{1 + \exp(-\xi)} = \frac{\exp(\xi)}{1 + \exp(\xi)}.$$

Logistic Regression

- Given $\{(\mathbf{x}_n, y_n) | n = 1, \ldots, N\}$, the likelihood is given by

$$
\begin{aligned}
p(\mathbf{y}|\mathbf{X}, \mathbf{w}) &= \prod_{n=1}^{N} p(y_n = 1|\mathbf{x}_n)^{y_n} (1 - p(y_n = 1|\mathbf{x}_n))^{1-y_n} \\
&= \prod_{n=1}^{N} \sigma\left(\mathbf{w}^\top \mathbf{x}_n\right)^{y_n} \left(1 - \sigma(\mathbf{w}^\top \mathbf{x}_n)\right)^{1-y_n}.
\end{aligned}
$$

- Then log-likelihood function is given by

$$
\begin{aligned}
\mathcal{L} &= \sum_{n=1}^{N} \log p(y_n|\mathbf{x}_n) \\
&= \sum_{n=1}^{N} \{y_n \log \sigma(\mathbf{w}^\top \mathbf{x}_n) + (1 - y_n) \log(1 - \sigma(\mathbf{w}^\top \mathbf{x}_n))\}.
\end{aligned}
$$

Softmax Regression

Softmax Regression

- Model input-output by a softmax transformation of logits $\theta_k = \mathbf{w}_k^\top \mathbf{x}_n$:

$$p(y_n = k|\mathbf{x}_n) = \mathrm{softmax}(\theta_k) = \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_n)}{\sum_{j=1}^K \exp(\mathbf{w}_j^\top \mathbf{x}_n)}.$$

- Given $\mathbf{Y} \in \mathbb{R}^{K \times N}$ and $\mathbf{X} \in \mathbb{R}^{D \times N}$, the likelihood is given by

$$p(\mathbf{Y}|\mathbf{X}, \mathbf{w}_1, \ldots, \mathbf{w}_K) = \prod_{n=1}^N \prod_{k=1}^K p(y_n = k|\mathbf{x}_n)^{Y_{kn}}.$$

- The log-likelihood is given by

$$\mathcal{L} = \sum_{n=1}^N \sum_{k=1}^K Y_{kn} \log(p(y_n = k|\mathbf{x}_n)).$$

Thank you.